FANAP INFRA IIoT Gateway

- Edge Gateway for Industrial IoT
- Main Card + Expansion Card





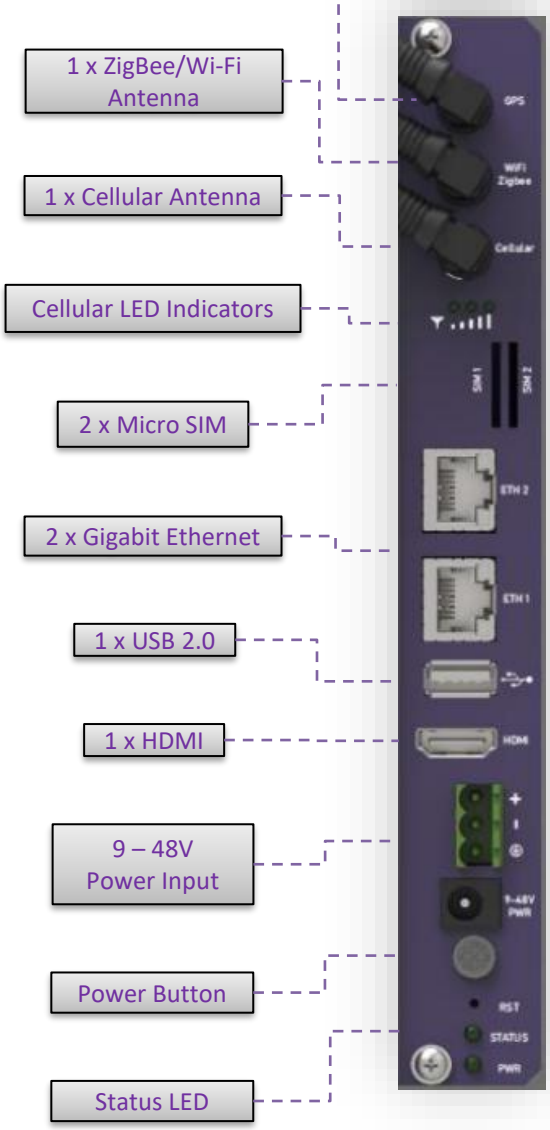
Introduction

- The FANAP INFRA Industrial IoT Edge Gateway, acts as a bridge between sensors, I/O devices, PLCs and the Internet, by collecting, processing, and standardizing data from industrial devices before sending to the cloud.

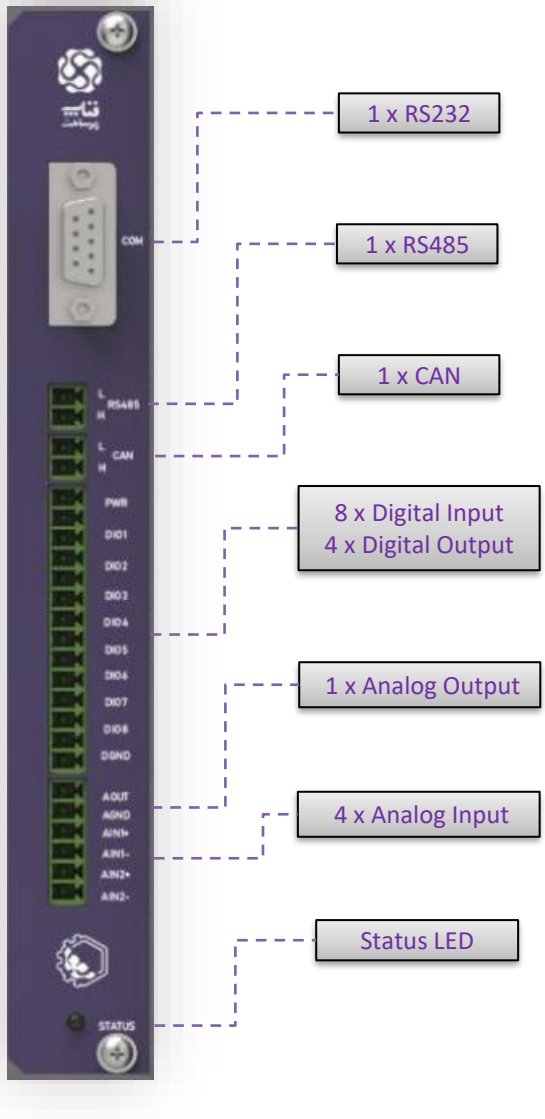
- The FANAP INFRA Gateway is a most configurable and manageable IIoT gateway that convert multiple protocols like MQTT, Modbus, S7, ZigBee, CoAP, and etc. to each others. real-time control, production and productivity monitoring and analysis of individual machines and entire plants, make it the great solution for industrial applications.

Plug-In Cards

Main Card



Expansion Card



❖ Key Features

- ✓ High performance, high reliability
- ✓ Flexible interfaces supporting numerous industrial applications
- ✓ Multi-protocol (MQTT, Modbus, s7 , ...)
- ✓ Seamlessly convert protocols for boundless machine-to machine/ machine-to-cloud data transmission and network connectivity
- ✓ Flexible and cost-effective design
- ✓ Providing Edge Computing
- ✓ Real-time performance
- ✓ Energy Management
- ✓ Quick customization meeting rapid market promotion
- ✓ Flexible Input Power
- ✓ Wide Temperature Range

Specifications

Operating System	Linux Kernel 4.14, Android Marshmallow	**Display**	HDMI 1.4A
Processor	TI AM3354, ARM Cortex-A8, up to 1GHz	**USB**	USB host 2.0 high speed
Memory	512MB DDR3L (supports up to 1GB)	**Serial Port**	• 2 x RS-485/422 • 1 x RS-232
Storage	• 4 GB eMMC onboard • alternatively 8/16/32/64 GB onboard flash memory	**CAN**	• 1 x Classic CAN 2.0 • 1 x CAN FD (Optional)
Cellular (Optional)	Standard: • GSM: 850/900/1800/1900MHz • UMTS: B1/B5/B8 • LTE FDD: B1/B3/B5/B7/B8/B20 • LTE/UMTS receive diversity	**Analog I/O**	4 x Analog Input: Channel-Independent Programmable Input Ranges: • Bipolar: ±10.24 V, ±5.12 V, and ±2.56 V • Unipolar: 10.24 V, 5.12 V
	Antenna Connector: SMA, female		1 x Analog Output: • Current output: 4 mA to 20 mA; 0 mA to 20 mA; 0 mA to 24 mA • Voltage output: 0 V to 5 V; 0 V to 10 V; ±5 V; ±10 V 0 V to 5.5 V; 0 V to 11 V; ±5.5 V; ±11 V (10% over range)
	SIM slot: Dual Micro SIM connector	**Digital I/O**	8 x Digital Input: • 24V Input Voltage
Ethernet	Port: 2 x RJ-45 (Eth1, Eth2)		4 x Digital Output: • 24V @300mA
	Speed: 2 x 10/100/1000 Mbps WAN	**Power Supply**	Connector: • 5.08 mm terminal block • 2.5 mm power jack
	Magnet isolation protection: 5KV		Input Voltage: 9 to 48V DC
ZigBee	Standard: IEEE 802.15.4		Input Current: 12V@3A, 24V@1.5A
	Frequency band: 2.4 GHz	**Power over Ethernet (PoE)**	Provides 1-port (Eth1) IEEE 802.3bt compliance PoE++, up to 90W
	Antenna Connector: SMA, female		Automatic Maintain Power Signature (MPS)
	Tx Output Power: • Option 1: +4.5 dBm (normal power) • Option 2: +22 dBm (high power)		Supports PoE++ PSE
	Rx Sensitivity: -98dBm	**Security**	Trusted Platform Module (TPM) v1.2 capable for Linux OS, Secure Boot, Security based on IEC 62443, Secure firmware update and I/O port disablement.
Wi-Fi	Standard: 802.11a/b/g/n	**Software**	Network protocols: TCP, UDP, DHCP, NAT, HTTP, HTTPs, DNS, RIP, OSPF, NTP, SMTP, Telnet, etc.
	Frequency band: 2.4 GHz		Serial port: TCP Client/Server, UDP, Modbus RTU Gateway
	Antenna Connector: SMA, female (combined with Bluetooth antenna)		Programming language: C, C++, Perl, Java
	Security: WEP, WPA, WPA2, WMM, WMM-PS (U-APSD), WMM-SA	**Mechanical and Environment**	Operating Temperature: -20℃～70℃
	Sensitivity: -90 dBm		Storage Temperature: -40℃～85℃
	Data Rate: up to 72.2Mbps		Dimensions: 115 * 77 * 22 mm (W * D * H)
Bluetooth	Standard: Bluetooth V4.0		Housing & Weight: Metal, 1.6 Kg
	Frequency band: 2.4 GHz		Installations: Desktop, VESA, DIN-rail and wall mounting
	Antenna Connector: SMA, female (combined with Wi-Fi antenna)	**Other Features**	1 x Onboard RTC (Real Time Clock)
	Sensitivity: -86 dBm		1 x Onboard Temperature Sensor
	Data Rate: up to 3Mbps		1 x Reset button
GPS	Antenna Connector: SMA, female		1 x Power button
	Supported antennas: Active and passive		LED indicators: • 1 x Status (Main control card) • 1 x Status (Expansion Card) • 3 x Modem Status • 1 x Link indicator + 1 x Speed indicator for each Ethernet port
	Protocols: NMEA, UBX binary, RTCM		
	Accuracy: 2.5m		
	Update Rate: 5Hz		
	Acquisition: Cold starts: 27 s ,Aided starts: < 3s, Hot starts: 1 s		
	Sensitivity: Tracking: –161 dBm, Cold starts: –147 dBm, Hot starts: –156 dBm		